

2	(a)	(resultant) force = rate of change of momentum / allow proportional to or change in momentum / time (taken)	B1	[1]
	(b)	(i) $\Delta p = (-) 65 \times 10^{-3} (5.2 + 3.7)$	C1	
		$= (-) 0.58 \text{ N s}$	A1	[2]
	(ii)	$F = 0.58 / 7.5 \times 10^{-3}$		
		$= 77(.3) \text{ N}$	A1	[1]
	(c)	(i) 1. force on the wall from the ball is equal to the force on ball from the wall but in the opposite direction (statement of Newton's third law can score one mark)	M1 A1	[2]
		2. momentum change of ball is equal and opposite to momentum change of the wall / change of momentum of ball and wall is zero	B1	[1]
	(ii)	<u>kinetic</u> energy (of ball and wall) is reduced / not conserved so inelastic (Allow relative speed of approach does not equal relative speed of separation.)	B1	[1]

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3 (a) metal:

regular / repeated / ordered arrangement / pattern / lattice